

3GPP TS 38.331

Radio Resource Control (RRC) protocol specification

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SYNTHETIC SAMPLE DOCUMENT — written for demonstration and testing of a retrieval-augmented chatbot. This is original content structured to resemble a 3GPP technical specification; it is NOT an official 3GPP publication and must not be used as a source of normative requirements.

4.2.1 RRC states

A UE in NR is, at any point in time, in one of three RRC states: RRC_IDLE, RRC_INACTIVE, or RRC_CONNECTED. In RRC_IDLE, the UE monitors paging and performs cell (re)selection but has no established RRC connection with any gNB. In RRC_INACTIVE, the UE retains its AS context and can resume the connection with reduced signalling overhead. In RRC_CONNECTED, a full RRC connection is established between the UE and the serving gNB, and the UE reports channel quality and mobility measurements as configured.

5.3.3 RRC connection establishment

RRC connection establishment is triggered by the UE for reasons including mobile originating data, mobile originating signalling, or a response to paging. The UE initiates the procedure by sending an RRCSetupRequest message on the common control channel. Upon receiving RRCSetupRequest, the gNB replies with RRCSetup, which configures Signalling Radio Bearer 1 (SRB1). The UE shall respond with RRCSetupComplete, after which the connection is considered established.

If the network is unable to admit the request, it shall respond with RRCReject and may include a wait timer instructing the UE to refrain from re-attempting for a specified duration.

5.3.13 RRC connection re-establishment

The UE shall initiate RRC connection re-establishment upon detecting radio link failure, upon handover failure, upon mobility from NR failure, upon integrity check failure, or upon an RRC connection reconfiguration failure, provided the UE has a stored AS security context and a suitable cell is found within the re-establishment timer T311.

6.3.2 Timers and constants (RRC)

The following timers govern RRC procedures. T300 bounds how long the UE waits for RRCSetup or RRCReject after sending RRCSetupRequest. T310 supervises detection of radio link failure following N310 consecutive out-of-sync indications. T311 bounds the UE's search for a suitable cell during re-establishment.

Timer	Default value	Started when	Stopped when
T300	1000 ms	RRCSetupRequest is sent	RRCSetup or RRCReject is received
T310	2000 ms	N310 out-of-sync indications received	N311 in-sync indications received
T311	3000 ms	Re-establishment procedure is initiated	A suitable cell is selected

6.3.5 Maximum values (RRC)

N310, the number of consecutive out-of-sync indications required before starting T310, and N311, the number of consecutive in-sync indications required to stop it, are both configurable by the network within the ranges below.

Constant	Range	Typical network setting
N310	1 to 20	1
N311	1 to 10	1